

Week 9 Final: Co-Design Canvas, Mock-Ups & System Requirements

Azaria Da Silva

Chosen Implementation: HCI:ED Triage Desk + Waiting Room Display

HCI (screen-based) was selected over HRI for this phase, as it extends the model already validated in Weeks 6-8 with no new hardware. Two connected screens are designed: a nurse-facing triage desk panel, and a public patient-facing waiting-room display.

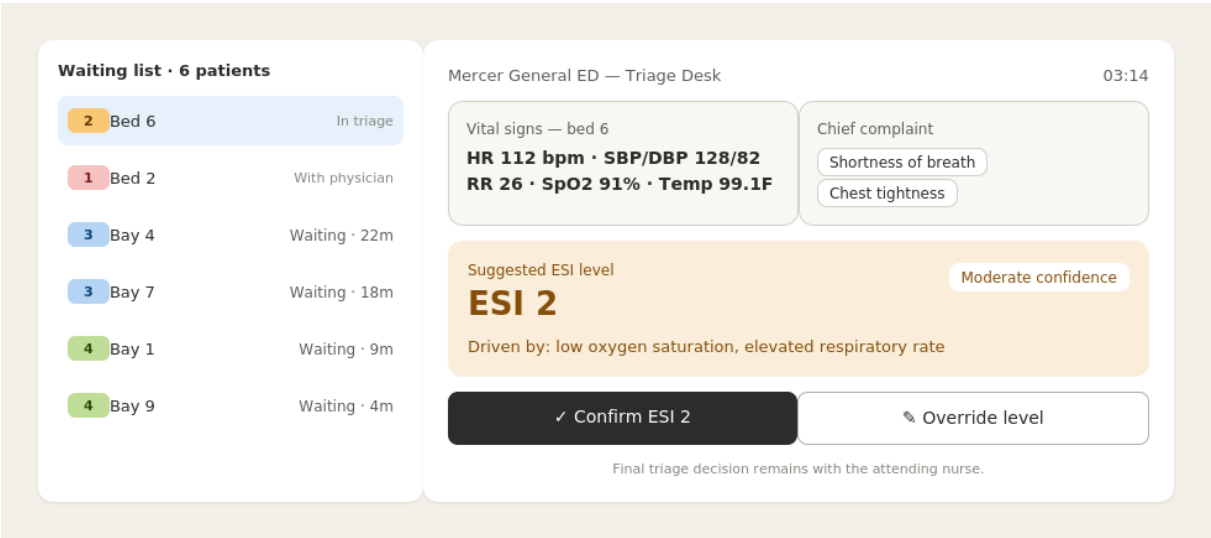
Completed Co-Design Canvas

Section	Content
Problem	Triage decisions currently vary by nurse and shift, and can be made before the full clinical picture is available. A screen-based decision-support tool at the ED triage desk can surface a model-suggested ESI level, with its reasoning, at the exact moment the nurse is deciding — without adding a new workflow step.
Ethics	The system must never auto-assign a triage level; the nurse retains final authority at all times. The interface must not create automation bias by presenting the suggestion more prominently than the nurse's own input fields. Confidence and top contributing factors are always shown alongside the prediction, never a bare number — and a low-confidence prediction visually de-emphasizes "Confirm" so the design itself does not nudge agreement. Any override is logged for later audit and model monitoring, but never blocked or discouraged. The patient-facing waiting display shows ticket numbers only — no names, ESI levels, or clinical detail are ever displayed publicly, protecting patient privacy and dignity in a shared waiting area.
Guidelines	Aligns to the five-level ESI standard already in clinical use — no new taxonomy introduced. Color choices follow WCAG AA contrast minimums and avoid red/green-only encoding for color-blind safety. The patient-facing display follows the same non-identifying-information principle used in public hospital ticketing systems generally. Hospital IT security and access-control guidelines (to be confirmed with Martina Griffith's team) will govern how the model connects to any live EHR feed.
MVP	Two connected screens: (1) a nurse-facing triage desk panel showing the patient queue, vitals, chief complaint, the model's ESI suggestion with a confidence label and short reasoning, and a one-click confirm/override control; (2) a public waiting-room display showing anonymized queue tickets, estimated wait by arrival band, and on-duty clinical staff availability. No robot or physical hardware component in this phase.
Environment	Main ED triage desk and adjoining waiting area, staffed and open continuously including overnight. High ambient noise, variable lighting, frequent interruption. The nurse-facing screen must be usable with brief, divided attention; the patient-facing display must be legible from several meters away in a shared public space.
Form	Software only — two screens within existing hospital display infrastructure. The nurse panel is a card-style layout embedded in the current triage desk software, sitting alongside rather than replacing existing tools. The waiting-room display is a wall-mounted or ceiling-mounted screen, styled distinctly (dark background, large type) for at-a-distance public legibility.

Mock-Ups

1. Nurse triage desk: queue and patient detail

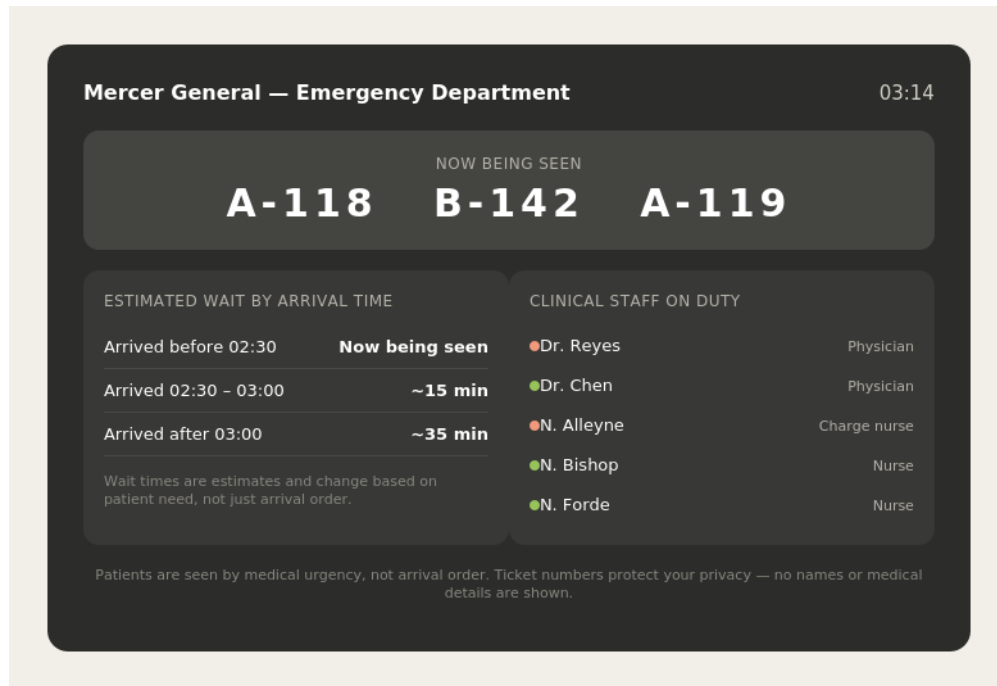
The primary clinical interface. Left panel shows the live patient queue with ESI color-coding and wait status; right panel shows the selected patient's vitals, chief complaint, and the model's suggestion with confidence and reasoning.



Nurse-facing triage desk: waiting list (left) plus vitals, chief complaint, ESI recommendation, confidence, and confirm/override controls (right).

2. Patient-facing waiting room display

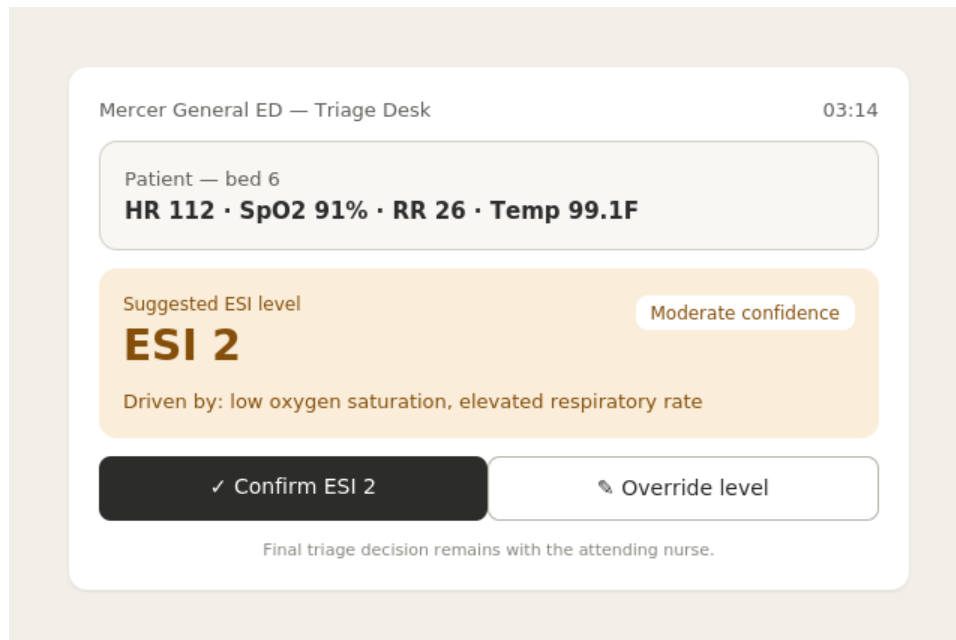
A public screen for the waiting area. Deliberately shows only anonymized ticket numbers and estimated wait bands; never names, ESI levels, or clinical detail, alongside which clinical staff are currently on duty and available.



Patient-facing display: now-serving tickets, estimated wait by arrival band, and on-duty staff availability. No patient-identifying or clinical information shown.

3. Nurse panel: confidence state comparison

Two contrasting states of the nurse panel, to test the design against both a confident and an uncertain prediction.



State A — moderate confidence. Confirm is the primary action.

Mercer General ED — Triage Desk
03:41

Patient — bed 3
HR 96 • SpO2 96% • RR 19 • Temp 98.4F

Suggested ESI level

⚠ Low confidence

ESI 3
Vitals within normal range; complaint pattern uncommon in training data
Rely on clinical judgement — this suggestion has weaker supporting evidence than usual.

✓ Confirm ESI 3

🛑 Override level

Final triage decision remains with the attending nurse.

State B — low confidence. Override, not Confirm, is the primary action, and an explicit caution note is shown — the interface never nudges agreement with an uncertain suggestion.

AI-Assisted Triage System Requirements Document

Category	Detail
Inputs (consumed)	Vitals (HR, SBP, DBP, RR, SpO2, temp, glucose) and chief-complaint flags, entered manually at triage as they are today. Queue position and staff duty roster are pulled from existing hospital scheduling systems, not the ML model. No demographic fields are consumed as model inputs, consistent with the Week 5-6 fairness decision.
Outputs (emitted)	Nurse panel: a predicted ESI level (1-5), a plain-language confidence label, and the top 2-3 contributing factors as short text. Waiting display: anonymized ticket numbers, an estimated-wait band by arrival time, and a live staff-availability roster (available / busy dot, role, no patient linkage).
What the human does with it	Nurse: reads the suggestion and reasoning, confirms or overrides in one click, proceeds with the existing workflow. Patient: reads their approximate wait band and sees which clinical staff are on duty — no action required, informational only.
Integration requirements	Model inference must complete in well under 1 second (consistent with the Week 6-7 benchmark showing millisecond-scale inference for logistic regression). Must degrade gracefully to a blank/neutral state, never a stale or misleading prediction, if the model service is unreachable. The waiting-room display must never surface any field the nurse panel does — a hard separation between clinician-facing and public-facing data is a system requirement, not a UI choice.
Non-functional requirements	Legibility: nurse panel readable under overnight low-light conditions; waiting display readable from 4+ meters. Privacy: patient-facing display contains zero personally identifying or clinical information by design. Auditability: every override is logged with a timestamp for later model monitoring, per the Week 7-8 decision-journal practice.

2-Minute Walk-Through Video Script

Script below, timed for approximately 2 minutes at a natural speaking pace (~130 words/min). Record as a screen-share walkthrough of the two mock-ups.

[0:00-0:20] The problem

"Hi, I'm walking through the design for our ED triage decision-support system. Today, triage decisions vary by nurse and by shift, and they sometimes get made before the full clinical picture is in. The goal here isn't to replace the nurse's judgement — it's to give them a second opinion, right at the moment they need it, using the model we validated in Weeks 6 through 8."

[0:20-0:50] The nurse-facing screen

"This is the triage desk panel. On the left, the live patient queue, color-coded by urgency. On the right, the selected patient's vitals and chief complaint — nothing new to enter, this is the same information already captured at triage. Below that, the model's suggestion: an ESI level, a confidence label, and the top factors driving it — not just a number. The nurse confirms or overrides in one click, and either way, the workflow continues exactly as it does today."

[0:50-1:20] Designing for uncertainty, not just the easy case

"Here's the same panel in two different states. When the model is confident, confirm is the primary button. But when the model is uncertain — vitals that don't clearly fit a pattern — the styling changes, a caution note appears, and Override becomes the primary action instead. This was a deliberate choice: the interface should never visually nudge a nurse toward agreeing with a prediction the model itself isn't sure about. That's directly tied to the automation-bias risk we flagged back in Week 4."

[1:20-1:45] The patient-facing display

"For the waiting room, we designed a second, public screen. It shows ticket numbers and estimated wait times, and which clinical staff are currently on duty — useful information for patients and families. But it deliberately shows nothing else: no names, no ESI levels, no clinical detail. That separation between what clinicians see and what the public sees is a hard system requirement, not just a UI choice."

[1:45-2:00] Close

"Both screens use the model we already built and validated — no new infrastructure, no new data streams. Next, we're looking forward to testing this design against a near-miss case, to see where it holds up and where it doesn't. Thanks for watching."